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BRIEF COMMUNICATION

Brief Separation Elevates Cortisol in Mother and Infant Squirrel Monkeys

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Received 27 August 1984

STANTON, M. E. AND S. LEVINE. *Brief separation elevates cortisol in mother and infant squirrel monkeys.* PHYSIOL BEHAV 34(6) 1007–1008, 1985.—The pituitary-adrenal response of mother and infant squirrel monkeys was assessed under basal conditions and after 5, 10, or 20 min of separation followed by 30 min of reunion. All separation conditions produced equal increments of cortisol (approximately 50%) in both mothers and infants. These increments were lower than would have occurred in the absence of reunion. These results extend previous reports that reunion ameliorates the psychoendocrine effects of separation and indicate that mother-infant attachment can be studied with shorter periods of separation than have been used customarily.

Mother-infant Separation Reunion Cortisol Squirrel monkeys

THE response of the pituitary-adrenal system has proven to be a valuable dependent measure for studying the process of attachment in nonhuman primates (see [1] for review). Attachment between mother and infant is evident both in the way maternal separation increases cortisol levels as well as in the way maternal reunion reduces, or inhibits increases in, cortisol levels [2]. Recently, it has been postulated that cognitive/learning processes may influence the adrenocortical and behavioral responses to separation stress in the squirrel monkey [4], as has been shown for other types of stress in a variety of species [3]. In future work, the role of learning processes within the separation paradigm could be studied by Pavlovian conditioning experiments in which maternal separation serves as an unconditioned stimulus (US). Such experiments could more easily be accomplished if US durations are shorter than 30 min, the minimum period of separation that has been shown to increase cortisol levels in mother and infant squirrel monkeys. Therefore, it is necessary to determine whether cortisol secretion can be increased by relatively brief periods of separation. Momentary separation does not substantially elevate cortisol levels after 30 min of reunion [5,6], whereas 30 min of separation does [2]. In order to further characterize the temporal dynamics of the adrenocortical response to separation, the present study examined the effects of different durations of separation between zero and 30 min. Mother-infant squirrel monkey dyads were separated for either 5, 10, or 20 min, allowed 30 min of reunion and then blood sampled.

METHOD

Animals

Six mother-infant pairs of Bolivian squirrel monkeys (*Saimiri sciureus*) were used in this study. The mothers were wild-born. The infants were laboratory-born and -reared, and were approximately 4 months of age at the start of the experiment. During the experiment, mother-infant pairs were housed in stainless steel Wahmann primate cages (64×74×68 cm). These cages were set in a temperature-controlled colony building having large windows which provided a natural light-dark cycle. All experimental manipulations occurred between 0900 and 1100 hr. Wayne Monkey Diet and water were continuously available. Fresh fruit and vitamin-enriched cherry drink were provided twice each week.

Procedure

The 6 mother-infant pairs were tested in three replications of 2 pairs each. The experiment began by transferring mother-infant pairs from group cages to individual cages where they were left undisturbed for 2–4 weeks. This was followed by four blood samples at weekly intervals. The first of these was a basal sample. Mother and infant were taken from the home cage and, within 3–4 min, were anesthetized with ethyl ether, and their blood was collected into heparinized syringes via cardiac puncture. The next three blood samples occurred immediately after a 30-min reunion period which was preceded by 5, 10, or 20 min of separation (the order in which these three separation intervals were encountered was counterbalanced across subjects to control for repeated sampling effects). Separation consisted of removing the infant from the home cage and placing it in a wire mesh cage (46×46×61 cm) situated within a sound-attenuating chamber in an adjacent room. The separation interval and the 30-min reunion period were timed with a stopwatch. Within each replication, only 1 of the 2 mother-infant pairs was separated on a given day. Further details of

the blood sampling procedure and the radioimmunoassay of cortisol in the squirrel monkey have been published previously [6].

RESULTS AND DISCUSSION

Cortisol levels (Fig. 1) were analyzed by means of a 2(Mother vs. Infant) × 4(Sample Condition) analysis of variance. This analysis revealed a main effect of Sample Condition, $F(3,30)=8.21$, $p<0.001$. No other main effects or interactions were statistically significant ($Fs<1$), indicating that the separation-induced increase in the mothers' cortisol did not differ from that of the infants. Post hoc tests (Newman-Keuls) of the main effect of Sample Condition indicated that all separation durations differed ($p<0.01$) from base but not from one another.

The findings of the present study indicate that separation periods as brief as 5 min elevate plasma cortisol in both mother and infant squirrel monkeys. This finding attests to the potency of the separation paradigm in activating the adrenal of mother-infant squirrel monkey dyads. Studies which examine the influence of learning factors on adrenocortical activation in infant squirrel monkeys can successfully use separation periods as short as 5 min, periods which are much briefer than those used in the past.

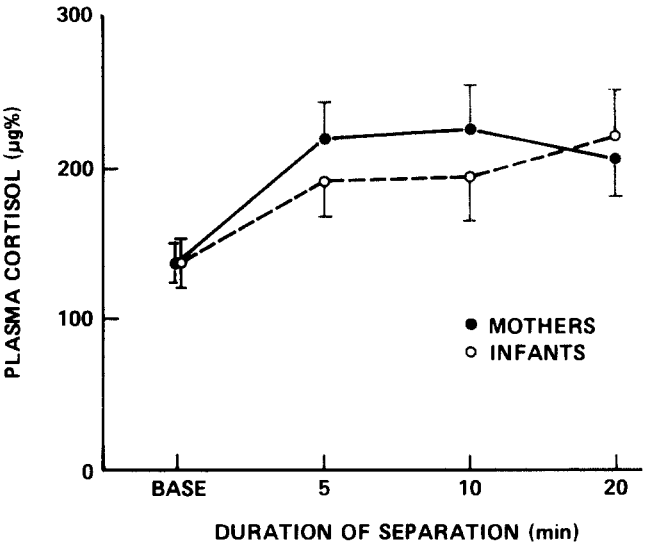


FIG. 1. Mean cortisol levels of mother and infant squirrel monkeys under basal conditions or after a 30-min period of reunion which was preceded by 5, 10, or 20 min of separation.

ACKNOWLEDGEMENTS

This research was supported by HD-02881 from NICH&HD, MH-23645 from NIMH, and Research Scientist Award MH-19936 from NIMH to S. Levine; and by Postdoctoral Training Grant MH-15147 from NIMH to M. Stanton. The authors thank Edna Lowe for technical assistance and Dr. Christopher Coe for criticisms of an earlier version of the manuscript.

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